



Bapuji Educational Association®
Bapuji Institute of Engineering and Technology, Davanagere
An Autonomous Institute Affiliated to VTU
Department of Computer Science and Engineering

Course Title	Database Management System	Course Code	BCSPCC403
Course Instructor	Dr. Anusha R, Prof. Ranjana B J, Prof. Kamala R, Prof. Deepa K B, Prof. Madhuri Martis	Semester-Section	IV
IA Test No.	I	Max. Marks	30
Date	24-04-2026	Time	9.30 am to 10.30 am

Course Outcome Statements: After the successful completion of the course, the students will be able to	
CO1	Conceptualize data modelling concepts to draw ER-diagrams for the given scenarios.
CO2	Create efficient queries to retrieve data from the database considered.
CO3	Apply the database normalization concepts to devise efficient databases for the given real time problems.
CO4	Analyze the importance of Transaction processing and concurrency control in efficient database management.
CO5	Differentiate between DBMS and NoSQL.

Note: Answer any one full question from each Part.					
Q. No.		Question	Marks	RBT Level	CO
		Part A			
1	A	What is database management system (DBMS)? Describe the characteristics of the database approach. Mention any two differences between DBMS and traditional file processing.	6	L2	1
	B	Define data model and data abstraction. Describe the different categories of data models.	6	L2	1
		OR			
2	A	Illustrate the three-schema architecture with a neat diagram. Briefly explain logical data independence and physical data independence.	6	L2	1
	B	Describe the types of attributes with an <u>example</u> . Also, Explain generalization and specialization with an example.	6	L2	1
		Part B			
3	A	The following relations are a part of the COMPANY database. EMPLOYEE (Name, Ssn, Bdate, Salary, Dno, Super_ssn) DEPARTMENT (Dname, Dnumber, Mgr_ssn, Mgr_start_date) PROJECT (Pname, Pnumber, Plocations, Dnum) DEPT_LOCATIONS (Dnumber, Dlocation) WORKS_ON (Essn, Pno, Hours) DEPENDENT (Essn, Dependant_name, Bdate, Relationship) Construct the following queries in relational algebra on the database schema: i) Retrieve the name of all employees who work for 'Research' department. ii) For every project located in 'Davanagere', list the project number and the department manager's name. iii) List the name of managers who have atleast one dependant.	6	L3	1
	B	Define the following with an example: (1) Primary key, (2) Foreign key, (3) Candidate key, (4) Super key (5) Domain, (6) Relation state	6	L2	1
		OR			
4	A	Identify and outline the referential integrity constraint violation in update operations on relational databases with suitable examples.	6	L2	1



Note: Answer any one full question from each Part.

Q. No.	Question	Marks	RBT Level	CO
B	Convert the following Entity Relationship (ER) diagram into relational database following the seven steps of ER to relational mapping.	6	L3	1
Part C				
5	A Analyze the following relational schema CUSTOMER (cid, cname, email, address, phone) ITEM (item_no, item_name, price, brand) SALES (cid, item_no, #items, amount, saledate) SUPPLIES (sid, sname, sphone, saddress) SUPPLY (sid, item_no, supplydate, qty) Develop the SQL queries for the following: i) List the items purchased by customer 'Prashanth'. ii) Get the details of customers whose total purchase of items worth is more than 5000 rupees. iv) List total sales amount, total items, average sale amount of all items.	6	L3	2
OR				
6	A Analyze the following relational schema SAILORS (sid, sname, rating, age) BOATS (bid, bname, color) RESERVE (sid, bid, day) Develop the SQL queries for the following: i) Retrieve the number of sailors whose age is greater than 50. ii) Retrieve the Sailor's name who has reserved boat number 103. iii) List the Sailor's name in ascending order.	6	L3	2

RBT (Revised Bloom's Taxonomy) Levels : Cognitive Domain

L1 : Remembering	L2 : Understanding	L3 : Applying
L4 : Analysing	L5 : Evaluating	L6 : Creating

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Course Instructor/ Coordinator

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Coordinator
DAAC

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HOD

BOE Chairman